

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Placebo-controlled Phase III Study to Evaluate the Efficacy, Safety, and Tolerability of Remibrutinib in Patients With Generalized Myasthenia Gravis, Followed by an Open-label Extension Phase
Brief Title: A Study to Investigate the Efficacy, Safety and Tolerability of Remibrutinib Versus Placebo in Adult Patients With Generalized Myasthenia Gravis **Short Title/Acronym:** RELIEVE
Protocol Number: CLOU064012301 **NCT Number:** NCT06744920 **Sponsor/Company Name:** Novartis Pharmaceuticals Corporation
Investigational Product: Remibrutinib (oral BTK inhibitor) **Phase of Development:** Phase III **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether treatment with remibrutinib will result in the reduction of the total score in the Myasthenia Gravis Activities of Daily Living (MG-ADL) scale as compared to placebo. **Secondary Objectives:** To evaluate the safety, tolerability, and long-term efficacy of remibrutinib in patients with generalized Myasthenia Gravis (gMG).

2.2 Background & Rationale To address the need for targeted, effective, and safe oral therapies in adult patients with Generalized Myasthenia Gravis (gMG) who remain symptomatic despite standard-of-care (SOC) treatment. Remibrutinib, a highly selective and potent covalent oral Bruton's tyrosine kinase (BTK) inhibitor, aims to improve daily living activities by targeting the underlying autoimmune mechanisms (B-cell activation and autoantibody production) driving the disease.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Core Part), Open-label (Extension Part) **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 180 Number of Sites: Multiple centers internationally **Estimated Study Start:** 07-Feb-2025 **Estimated Primary Completion:** 26-Feb-2033 (Overall Completion)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients with gMG (age 18-75 years). 2. Confirmed diagnosis of Myasthenia Gravis Foundation of

America (MGFA) Class II-IV gMG at screening and likely not in need of a respirator for the duration of the study, as judged by the Investigator. 3. Documented evidence of positive serologic testing for AChR+ antibody or MuSK+ antibody at screening, OR seronegative for both AChR and MuSK antibodies at screening. 4. Baseline MG-ADL score ≥ 6 with $\geq 50\%$ of the total score due to non ocular symptoms. 5. Participants who have been on a stable dose of standard-of-care treatment as specified in the protocol. 6. Able to safely swallow the study medication according to investigator clinical judgement based on a bedside swallowing test or another formal swallowing test in line with local practice, both at Screening and Baseline.

4.2 Exclusion Criteria 1. Women of child-bearing potential, defined as all women physiologically capable of becoming pregnant, unless they are using highly effective methods of contraception during dosing and for 1 week after stopping of study treatment. 2. Prior to baseline have been treated with intravenous immunoglobulins or plasma exchange (IVIg/PLEX) in the past month, with rituximab in the past 6 months, eculizumab in the past 2 months, ravulizumab or other complement inhibitors in the past 3 months, efgartigimod or other anti-FcRn therapies in the past 3 months, or had a thymectomy in the past 6 months or a planned thymectomy during the trial period. 3. Severe uncontrolled medical conditions or active infections outside of the primary gMG diagnosis. 4. Recent use of prohibited immunosuppressants or targeted biological therapies that are not part of the stable SOC protocol.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral film-coated tablet, continuous daily dosing. **Route of Administration:** Oral **Duration of Treatment:** 6 months (double-blind Core Part) followed by up to 60 months (open-label Extension Part).

5.2 Concomitant & Prohibited Therapy Permitted: Stable standard-of-care (SOC) treatment for gMG. **Prohibited:** Investigational treatments or newly initiated immunomodulatory therapies outside of the standard trial guidelines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline MG-ADL assessment
Baseline	Day 0	Randomization, First Dose, Vital Signs
Interim 1	Month 3 $\pm$ 7	Safety Labs, Efficacy Assessment (MG-ADL), AE Review
End of Core	Month 6	Primary Efficacy Assessment, Option to enter Open-Label Extension
End of Study	Month 66	Final Vital Signs, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Change from baseline to Month 6 in Myasthenia Gravis Activity of Daily Living (MG-ADL) total score. *Measurement Method:* Clinical questionnaire/assessment (MG-ADL scale).

7.2 Secondary & Exploratory Endpoints 1. Change from baseline to Month 6 in Quantitative Myasthenia Gravis (QMG) total score. 2. Proportion of participants with ≥ 5 points reduction from baseline to Month 6 of QMG total score without rescue medication and/or strongly confounding prohibited medication. 3. Proportion of participants with ≥ 3 points reduction from baseline to Month 6 of Myasthenia Gravis Activity of Daily Living (MG-ADL) scale total score without rescue medication and/or strongly confounding prohibited medication. 4. Incidence and severity of Adverse Events (AEs) to establish the safety and tolerability profile. 5. Long-term change in MG-ADL score and symptom control over the 60-month open-label extension.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient reporting and clinical assessment during scheduled visits.

Serious Adverse Events (SAEs): Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External independent Data Monitoring Committee to evaluate ongoing safety and tolerability.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for adverse events evaluations. **Sample Size Justification:** Target $N=180$ appropriately powered ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect a statistically significant reduction in MG-ADL scores versus the placebo group. **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) or Multiple Imputation for longitudinal continuous MG-ADL score data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol compliance and data integrity. **Audit Trail:** Electronic Case Report Form (eCRF) locking and resolution via standard data clarification forms.

11. Study Identifiers & Governance

Primary ID: CLOU064012301 **Registry Number:** NCT06744920 **Sponsor/Lead Organization:** Novartis Pharmaceuticals Corporation **Study**

Title: RELIEVE **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled Phase III Study to Evaluate the Efficacy, Safety, and Tolerability of Remibrutinib in Patients With Generalized Myasthenia Gravis, Followed by an Open-label Extension Phase

12. Clinical Framework (gMG Specific)

Primary Condition / Disease Area: Generalized Myasthenia Gravis (gMG) **Diagnosis Threshold:** MG-ADL score ≥ 6 (>50% non-ocular) **Medical Methodology:** Stable standard-of-care (SOC) treatment + targeted BTK Inhibition **Optimization Target:** Clinically meaningful reduction in MG-ADL score and symptom control

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for gMG symptom tracking **Education Protocol (HCP):** Protocol training and MG-ADL assessment certification **Education Protocol (Patient):** Onsite training regarding study drug administration, double-blind compliance, and symptom reporting **Quality Audit Mechanism:** Site monitoring and centralized review of clinical assessments

14. Observation & Follow-Up Schedule

Total Duration: Up to 66 months (6-month core + 60-month extension) **Onsite Visit Cadence:** Baseline, Month 3, Month 6, and specified intervals during the extension phase **Remote Monitoring Frequency:** Every 4 weeks (Q4W) or as specified by the protocol **Checklist Review Interval:** Every 12 weeks (Q12W) for continuous safety review

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				